

Tri Pham

p.tri@wustl.edu • Saint Louis, MO 63130 • (785) 331-6501 • [linkedin.com/in/vothientripham](https://www.linkedin.com/in/vothientripham)

SUMMARY

- PhD candidate in Computer Science with published research in Journal of Computer Languages and IEEE VL/HCC Conference on developer tools, code history interfaces, and interruption recovery in software development.
- 4+ years of HCI research experience designing and conducting within-subjects user studies, including development of integrated code capture frameworks for VS Code and Chrome recording fine-grained programmer activities.
- Proven mentor and educator with experience giving class lectures, supervising graduate/undergraduate thesis research, and supporting students in technical coursework.
- Strong technical proficiency in Python, JavaScript, C++, and extension APIs with expertise in usability testing, think-aloud protocols, and user-centered design methodologies.
- International scholar with demonstrated leadership across research, teaching, and residential life roles.

EDUCATION

Washington University in Saint Louis, McKelvey School of Engineering | Saint Louis, MO

Master of Science in Computer Science

May 2024

Doctor of Philosophy in Computer Science, Advisor: Dr. Caitlin Kelleher

May 2027

The University of Kansas (KU), School of Engineering | Lawrence, KS

May 2021

Bachelor of Science in Computer Science, Minor: Mathematics, Graduated with Honors

GPA: 3.91

SKILLS

Instrumentation: Python, JavaScript, C++, Extension APIs [Visual Studio Code (VS Code), Google Chrome, OpenAI]

Techniques: Usability Testing, Think-Aloud Protocols, Contextual Inquiry, Interviews and Surveys, Task Analysis, User-Centered Design, Iterative Design, Prototyping, Heuristic Evaluation, Cognitive Walkthrough, A/B Testing, Log Data Analysis

Software: Visual Studio Code, PyCharm, Git, GitHub, Microsoft Office, Google Products, Zoom Workplace, Slack

Languages: Vietnamese (Native) and English (Fluent)

RESEARCH EXPERIENCE

Graduate Research Assistant, Dr. Caitlin Kelleher's Lab

September 2022 – Present

Washington University in Saint Louis, Department of Computer Science & Engineering | Saint Louis, MO

- **Developed and maintained an integrated code capture framework** for VS Code and Chrome that records fine-grained development activities (code edits, executions, web resources) to enable detailed analysis of programmer behavior and code evolution.
- Designed and conducted 28+ within-subjects user studies investigating how different interface designs (chronological and sub-goal-based history) support developers' context recovery strategies and **formalizing recovery as multi-stage process model rather than a single event**.
- **Investigating the relationship between developers' information needs and interruption duration, and exploring LLM-supported interfaces to answer questions about code base history** (why was the code changed this way, how does it implement this behavior) using captured development context.
- **Mentored a student through their academic progression from undergraduate senior to master's degree**, advising on thesis projects regarding developer tools and interruption recovery (Fall 2024 – Summer 2025).
- Served as Graduate Teaching Assistant in a Human-Centered Design coursework (Fall 2025) and **delivered lectures on Cognitive Load Theory and its applications in interface design**.

Research Intern, Dr. Gerald Schweiger's Lab

Summer 2020

Graz University of Technology (TU Graz) | Graz, Austria

- Collaborated with peer researchers to synthesize state-of-the-art findings in building energy simulation.
- Assisted in the development and validation of different methods to forecast the Engineering building energy usage.
- Researched feature selection and relevant models for building energy consumption prediction.
- Tested hyperparameter tuning and implemented visualization for statistical models.

PUBLICATIONS

Vo Thien Tri Pham and Caitlin Kelleher. 2025. Code histories: Documenting development by recording code influences and changes in code. Journal of Computer Languages 82 (March 2025), 101313. DOI: <https://doi.org/10.1016/j.col.2024.101313>.

Vo Thien Tri Pham, Haixin Zhou, and Caitlin Kelleher. 2025. Interruptions and Recovery: Leveraging Dynamic Code History in Development. 2025 IEEE Symposium on Visual Languages and Human-Centric Computing. Accepted October 2025.

PRESENTATION

Presenter, IEEE Symposium on Visual Languages and Human-Centric Computing (VL/HCC)
Duke Energy Hall, North Carolina State University, Raleigh, NC

October 2025

WORK EXPERIENCE

Supplemental Instructor

August 2020 – May 2021

KU School of Engineering, Supervisor: Dr. David Johnson | Lawrence, KS

- Reviewed targeted supplemental materials by analyzing lecture content, problem sets, and exam patterns to address common student challenges and misconceptions.
- Helped 50 students each semester with questions about in-class problems, reinforcing complex machine learning concepts including supervised/unsupervised learning, neural networks, and model evaluation techniques.
- Provided one-on-one coding assistance for Python-based programming assignments, debugging algorithms and helping students develop computational problem-solving skills.
- Tracked student progress and communicated recurring challenges to Dr. Johnson to inform instructional adjustments.

Resident Assistant

April 2020 – May 2020

KU Jayhawker Towers | Lawrence, KS

- Rapidly transitioned from Food Board Manager at Pearson Scholarship Hall to Resident Assistant at Jayhawker Towers during COVID-19 pandemic to support students remaining on campus.
- Maintained consistent student support presence during university-wide emergency operations, doing patrol rounds and being on-call to respond to lockouts, room conflicts, and resident concerns.
- Enforced CDC and Kansas Department of Health COVID-19 safety protocols while balancing student well-being in an environment with suspended guest policies and restricted communal spaces.

Food Board Manager

August 2019 – April 2020

KU Pearson Scholarship Hall | Lawrence, KS

- Oversaw \$100,000 food budget during the current academic year period; when COVID-19 pandemic happened, the remaining well-budgeted amount served as necessary relief funds for impacted students.
- Ensured that all dietary needs are appropriately met, and all food was adequately stocked.
- Worked alongside a peer Proctor and reported to a Graduate Resident Assistant to support smooth hall operations.

Public Service Intern

May 2018 – May 2019

KU Honors Program | Lawrence, KS

- Developed a comprehensive listing of previous members of the Honors Program as a part of a History Project.
- Compiled information and documented it in a quantitative manner.
- Assumed special assignments and other duties as designated by the Director of Advancement.

Orientation Assistant

March 2018 – April 2019

KU First-Year Experience (FYE) | Lawrence, KS

- Assisted Class of 2022 students with their academic, social, and personal transition to KU.
- Served as ambassadors for other FYE programs including Hawk Week, KU Common Book, and Seminars.
- Developed positive relationships and maintained open communication with students and guests at orientation programs and throughout students' first year at KU.

SCHOLARSHIPS & AWARDS

- | | |
|--|-------------------------|
| • KU International Excellence Award (Undergraduate Full Tuition Scholarship) | Fall 2017 – Spring 2021 |
| • KU Honors Opportunity Award | Summer 2020 |
| • Dr. Robert C. and Lois E. Adams Nelson Scholarship | Fall 2019 – Spring 2020 |
| • Carol and Paul Gerlach Scholarship | Fall 2018 – Spring 2019 |
| • Garmin Excellence Scholarship | Fall 2018 – Spring 2019 |
| • Garmin EECS Scholarship | Fall 2018 – Spring 2019 |
| • 2nd Place, 36th Annual Mathematics Prize Competition, KU Math Department | April 2018 |